Science Teacher, 8A

Vasant Valley School, New Delhi

Subject: Science

3139/S8A/0018

We are pleased to present the ASSET Teacher MyBook, a comprehensive resource designed to provide you with valuable insights into your students performance and support their academic growth. This resource offers a range of features to help you better understand your students' strengths and areas for development.

Personalized Teacher Feedback: Gain tailored insights into your students' distinctive strengths and areas for development. This feedback is aimed at assisting you in providing targeted guidance to each student.
Student Performance Table: Delve into the individual question-wise performances of your students. This data-driven view will aid you in identifying specific areas where students may require additional support.
Bird's Eye View: Acquire a holistic comparison of your students' performance against the national average. This perspective will help you contextualize your students' achievements and progress.
Year on Year Analysis - Previous Batch: This report compares student performance over a year.It compares the performance of a specific class compared to the previous year.
Skill-based Summary: Explore a tabulated display of students' skill proficiency, emphasizing the skills with the highest and lowest performance. This summary enables you to gauge skill performance relative to the school and national average.
Analysis of Easy/Difficult Questions: Identify questions posing ease or challenge compared to the national norm. This section illuminates areas needing focused attention, allowing you to tailor your teaching approach accordingly.
Questions with Common Wrong Answer: This section highlights potential misconceptions through selected questions. It provides valuable insights into your students' performance within a broader nationwide context.
Practice Questions and Explanation: In the Student MyBook, students receive practice questions tailored to their needs. This Teacher MyBook also features practice questions customized for your class, complete with answers and explanations. Different practice approaches have been thoughtfully curated based on subject and skill requirements.

We recommend that you review this analysis with reference to the ASSET question paper. Your commitment to your students' growth is highly commendable, and we believe that the ASSET Teacher MyBook will further empower you in this journey.

Should you have any questions or require further assistance, please feel free to reach out to us at info@ei.study. We are dedicated to supporting your efforts in enhancing your students' learning experience.

Regards,
The ASSET Team

Kindly note: Some images may slightly appear unclear due to technical reasons, but rest assured, this is not an error.

Dear Science Teacher,

Congratulations on receiving your students' ASSET test results!

ASSET helps us understand where your students excel and where they might need more support.

Let's celebrate the strengths your students have shown in Science are:

Advanced or complex data representation or interpretation
Integrating different concepts or information for decision making

Great job nurturing these talents!

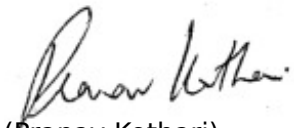
Now, let's look at areas where we can help them grow even more. Here's where they can improve in Science :

Recollection or recognition of science facts and concepts
Extraction, translation and application of knowledge or information

To assist you in this journey, we've created a special booklet. It's designed to focus on the questions/ skills that need a little boost. The booklet provides insights into each student's progress and highlights questions that sparked interest or posed a challenge.

Take your time going through the analysis to understand how each student is doing. You'll find different sections in the booklet to spot learning gaps and dive into specific questions that need attention. Your students have personalized booklets, called Student MyBooks, too. These are tailored to their needs, giving them a glimpse into the areas where they can practice more.

Feel free to use these resources in class, and let us know how it goes. We're here to help and learn from your experience. Share your thoughts with us at info@ei.study.As you guide your students through their learning journey, you're helping them become even better. Best of luck!



(Pranav Kothari)
Chief Executive Officer - Educational Initiatives

In case of queries, please write to us at info@ei.study or contact our customer care executive at the toll free number 1800-102-8885 (9 am - 6 pm, Mon - Fri).

No.	Student	Question No: →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	Score	Scaled Scores	Percentile	Performance
		Correct Answers →	D	B	A	D	A	B	C	D	D	B	A	C	A	C	D	D	B	C	A	A	C	D	B	B	A	B	C	B	A	D	B	D	A	A	B	A	D	B	C	C	D	C	A	D	A				
1	Aadvya Manhas		B	C	✓	C	✓	✓	D	✓	C	✓	✓	B	✓	✓	C	✓	D	✓	D	D	✓	C	✓	D	✓	✓	✓	D	✓	C	✓	C	✓	B	✓	D	A	C	✓	✓	B	B	✓	✓	D	23	548	72	
2	Aarav Chauhan		B	✓	✓	C	D	A	A	C	C	C	✓	✓	D	B	A	C	D	✓	B	✓	✓	C	D	C	D	D	A	✓	✓	C	✓	A	✓	C	C	D	B	D	A	B	C	A	B	✓	✓	13	429	25	
3	Alisya Kalha		C	C	C	C	B	A	✓	B	✓	A	✓	✓	D	✓	A	C	D	B	✓	B	✓	A	✓	C	✓	✓	A	✓	✓	✓	✓	✓	✓	✓	✓	D	C	A	B	✓	A	✓	B	✓	B	21	564	77	
4	Amaira Shah		✓	✓	C	A	C	A	✓	A	✓	C	✓	D	✓	B	C	A	D	B	✓	D	✓	✓	D	D	B	C	A	D	✓	C	✓	✓	✓	✓	✓	D	✓	C	A	B	✓	✓	✓	✓	B	20	551	73	
5	Anshu Balmiki		C	✓	✓	A	C	✓	D	B	C	✓	✓	B	D	B	C	A	D	✓	✓	B	✓	A	✓	C	B	D	A	C	B	✓	A	✓	C	B	C	B	A	D	B	✓	C	B	✓	C	B	13	389	10	
6	Arjun Sardana		C	C	C	A	D	D		B	C	C	✓	A	D	A	B	C	D	B	B	B	D	B	C	C	D	A	D	✓	✓	A	✓	A	B	C	C	D	A	A	A	D	A	A	✓	C	B	5	314	0	
7	Aryaveer Singh		✓	✓	B	C	✓	✓	✓	C	✓	✓	B	✓	✓	✓	B	✓	✓	D	B	✓	✓	A	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓	B	✓	✓	✓	C	✓	✓	✓	✓	B	✓	✓	33	742	99	O
8	Avni Garg		C	✓	C	C	C	✓	D	B	✓	✓	D	✓	D	✓	B	C	C	A	C	C	D	C	✓	✓	D	✓	A	A	✓	B	✓	B	C	✓	✓	D	C	✓	B	✓	A	✓	B	✓	✓	18	495	52	
9	Gargi Malik		C	✓	C	C	C	✓	D	B	✓	A	B	✓	D	A	C	C	✓	A	B	C	D	C	✓	C	✓	✓	A	*	✓	B	C	A	B	D	C	D	C	✓	B	✓	A	D	C	✓	✓	13	420	21	
10	Inaaya Irfan		B	✓	*	B	✓	C	✓	C	✓	✓	✓	✓	✓	B	B	✓	C	A	C	B	*	*	C	✓	D	C	✓	✓	✓	✓	✓	✓	✓	✓	✓	D	C	✓	✓	D	C	D	✓	✓	✓	24	565	77	
11	Jaiveer Bhutani		C	D	C	C	✓	✓	✓	C	✓	✓	B	✓	✓		B	*	✓	D	B	✓	✓	*	✓	*	✓	✓	✓	✓	✓	B	✓	✓	✓	B	✓	✓	✓	C	✓	✓	✓	✓	B	✓	✓	28	652	93	
12	Katha S Sahib		C	C	C	C	B	C	✓	B	✓	A	C	✓	✓	✓	A	A	D	D	D	✓	✓	C	✓	D	B	✓	A	D	✓	✓	✓	A	✓	B	✓	D	B	✓	B	✓	A	A	B	✓	D	17	509	58	
13	Kian Jaruhar		C	C	D	B	C	A	✓	B	B	A	D	B	D	✓	B	A	A		D	D	D	✓	C	D	B	D	A	C	✓	✓	D	A	✓	B	A	✓	A	C	A	B	✓	✓	✓	B	B	10	399	13	
14	Kuvira Kumar		C	C	✓	B	*	C	✓	C	✓	✓	✓	B	D	✓	A	B	C	✓	D	B	✓	✓	✓	✓	✓	C	✓	✓	✓	C	C	✓	✓	B	✓	D	C	A	B	✓	✓	✓	B	✓	✓	23	546	71	
15	Kyra Wirk		C	✓	C	B	C	A	A	A	✓	✓	✓	✓	✓	✓	B	A	D	D	C	C	✓	✓	✓	✓	D	✓	✓	✓	✓		✓	B	✓	B	✓	D	✓	A	B	✓	A	✓	B	B	✓	22	565	77	
16	Mayank		A	C	C	B	D	✓	B	C	C	✓	D	A	C	✓	A	B	C	B	C	B	✓	C	✓	D	B	D	B	D	✓	C	✓	C	✓	B	A	C	C	✓	B	D	B	A	C	C	B	9	394	12	
17	Nathania Chingsubam		✓	✓	C	✓	C	✓	A	C	✓	✓	✓	✓	✓	✓	B	C	D	B	✓	B	✓	C	C	✓	D	✓	A	C	✓	C	✓	C	B	✓	A	D	✓	A	✓	✓	A	D	B	✓	✓	22	565	77	
18	Nikitha S Muthappa		✓	✓	C	✓	C	✓	✓	A	✓	✓	✓	✓	✓	✓	B	C	✓	✓	B	B	✓	✓	C	*	D	✓	A	✓	✓		✓	A	✓	B	✓	✓	B		B	✓	✓	✓	D	✓	✓	27	658	94	
19	Reanna Malik		C	✓	C	A	C	A	✓	✓	✓	C	B	✓	✓	✓	C	C	✓	✓	B	C	D	✓	C	D	D	A	B	A	C	B	A	B	✓	C	D	D	A	A	B	✓	A	A	B	✓	✓	14	425	23	
20	Riddhhima Sahu		C	✓	C	B	C		✓	A	✓	C	✓	B	✓	✓	B	C	D	✓	B	✓	D	✓	C	A		✓	✓	D	✓	C	✓	A	✓	B	✓	D	B	A	✓	B	A	✓	B	✓	B	18	516	61	
21	Rudra Madan		A	✓	C	A	D	✓	✓	A	C	A	B	✓	✓	✓	B	✓	A	✓	B	D	✓	✓	✓	C	C	✓	A	✓	✓	C	✓	C	✓	B	C	D	✓	C	B	B	A	✓	B	✓	B	19	539	69	
22	Ruhaan Singh		B	✓	✓		D	A	✓	C	C	C	✓	B	✓	✓	C	A	✓	✓	D	B	✓	C	✓	✓	D	D	A	✓	✓	C	✓	A	✓	✓	✓	C	✓	✓	✓	✓	A	✓	B	✓	✓	24	571	79	
23	Satya V Gupta		✓	✓	C	✓	✓	D	✓	B	✓	✓	✓	✓	✓	✓	A	✓	D	A	B	B	✓	✓	✓	C	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	✓	D	C	A	✓	✓	A	✓	B	✓	✓	30	686	96	
24	Shoaib Ahmad		B	✓		A	D	A	✓	B	C	C	✓	B	✓	B	C	A	✓	✓	D	B	✓	C	✓	✓	D	A	A	✓	✓	B	✓	A	✓	✓	✓	C	✓	✓	✓	✓	A	✓	B	✓	✓	22	547	72	
25	Shrey Gujral		✓	✓	B	C	✓	✓	✓	C	✓	✓	B	✓	✓	✓	B	✓	✓	D	B	✓	✓	A	✓	✓	✓	✓	✓	✓	✓	B	✓	✓	✓	B	✓	✓	✓	C	✓	✓	✓	✓	B	✓	✓	33	742	99	O
26	Sifath Lohia		C	C	B	A	C	D	A	B	C	A	C	D	D	B	A	C	✓	B	D	✓	✓	C	✓	A	D	✓	A	A	C	✓	✓	✓	✓	C	✓	✓	✓	D	✓	✓	✓	B	C	A	✓	16	448	32	
27	Soham Jha		✓	D	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	C	✓	✓	✓	✓	✓	C	✓	✓	D	✓	✓	✓	✓	C	✓	A	✓	C	✓	✓	✓	D	✓	✓	✓	B	✓	✓	✓	35	745	99	O
28	Sophie Singh			✓	C	B	C	A	A	A	✓	✓	✓	✓	✓	✓	B	A	D	D	C	C	✓	A	✓	✓	D	✓	✓	✓	✓	C	✓	B	✓	B	✓	D	✓	A	B	✓	A	✓	B	B	✓	21	551	73	
29	Vani Garg		B	✓	✓	B	✓	C	✓	C	✓	✓	✓	✓	✓	✓	A	C	D	✓	✓	B	✓	✓	✓	✓	D	C	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	D	✓	✓	B	✓	✓	33	712	98	
30	Vani Verma		A	✓	C	B	✓	A	B	✓	✓	A	✓	D	D	✓	B	B	D	D	*	C	*	*	✓	✓	B	D	✓	D	✓	✓	✓	B	✓	D	A	B	A	✓	A	B	*	✓	✓	B	D	16	455	36	

% answered correct (section)	27	67	21	12	33	42	64	15	67	52	64	64	64	73	0	24	30	36	18	27	73	36	70	42	30	58	42	58	91	27	85	39	85	27	67	27	48	27	45	61	33	58	27	76	61	545.2/500			
% answered correct (school students)	31	68	12	15	30	58	64	8	65	46	61	57	55	81	6	27	38	45	18	18	72	52	57	38	22	56	48	50	92	31	81	35	79	34	52	30	42	20	43	56	36	60	27	70	56	539.9/500			
% answered correct (all students)	41	59	26	44	39	58	54	12	50	41	53	43	27	75	8	35	45	44	26	27	68	50	56	35	31	48	40	37	80	32	67	28	62	36	39	34	33	27	33	40	34	51	26	55	45	505.5/500			

Notes: Score: Actual score obtained by the student. **Percentile:** Percentage of students who have scored less than that particular student. **Performance:**'O' indicates outstanding performance in subject.
✓ Indicates correct answer. **A,B,C,D:** Indicates student's incorrect answer. **(Blank)** indicates student omitted answer. ***** Indicates student chose more than one option.
Scaled Score: Raw scores are converted to a scaled score through a process known as equating. The minimum a student can get is 200 and the maximum is 800.

		Question No: →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	Score	Scaled Scores	Percentile	Performance
No.	**Student↓**	**Correct Answers →**	**D**	**B**	**A**	**D**	**A**	**B**	**C**	**D**	**D**	**B**	**A**	**C**	**A**	**C**	**D**	**D**	**B**	**C**	**A**	**A**	**C**	**D**	**B**	**B**	**A**	**B**	**C**	**B**	**A**	**D**	**B**	**D**	**A**	**A**	**B**	**A**	**D**	**B**	**C**	**C**	**D**	**C**	**A**	**D**	**A**				
31	Veer Kumar		✓	✓	B	C	C	✓	✓	A	A	D	B	✓	✓	✓	B	✓	C	A	C	✓	*	✓	D	D	✓	✓	B	C	✓	C	✓	✓	✓	✓	C	D	✓	C	A	B	✓	✓	✓	✓	B	22	563	76	
32	Veer Ramchandani		C	D	C	C	✓	C	✓	✓	✓	✓	✓	✓	✓	✓	A	A	D	D	C	C	✓	C	✓	✓	✓	D	✓	✓	✓	C	✓	A	✓	B	✓	✓	✓	A	✓	B	B	D	B	✓	✓	24	581	81	
33	Viraaj Kathuria		✓	✓	C	B	✓	✓	✓	*	✓	C	✓	✓	C	✓	B	✓	D	D	D	B	✓	✓	✓	C	D	✓	A	✓	✓	C	✓	✓	✓	C	✓	D	✓	A	✓	B	B	B	C	✓	D	23	606	87	

% answered correct (section)	27	67	21	12	33	42	64	15	67	52	64	64	64	73	0	24	30	36	18	27	73	36	70	42	30	58	42	58	91	27	85	39	85	27	67	27	48	27	45	61	33	58	27	76	61	545.2/500			
% answered correct (school students)	31	68	12	15	30	58	64	8	65	46	61	57	55	81	6	27	38	45	18	18	72	52	57	38	22	56	48	50	92	31	81	35	79	34	52	30	42	20	43	56	36	60	27	70	56	539.9/500			
% answered correct (all students)	41	59	26	44	39	58	54	12	50	41	53	43	27	75	8	35	45	44	26	27	68	50	56	35	31	48	40	37	80	32	67	28	62	36	39	34	33	27	33	40	34	51	26	55	45	505.5/500			

Notes: Score: Actual score obtained by the student. **Percentile:** Percentage of students who have scored less than that particular student. **Performance:**'O' indicates outstanding performance in subject.
✓ Indicates correct answer. **A,B,C,D:** Indicates student's incorrect answer. **(Blank)** indicates student omitted answer. ***** Indicates student chose more than one option.
Scaled Score: Raw scores are converted to a scaled score through a process known as equating. The minimum a student can get is 200 and the maximum is 800.

The Bird's Eye View shows a comparison of the performance of your students with the national average performance.

Your class 8 A, has performed better than the national average in the Science ASSET paper this year.

Class 8	Science	No. of students
8A (Your Class)	109%	33
8B	107%	31
8C	107%	34
All sections combined	108%	98

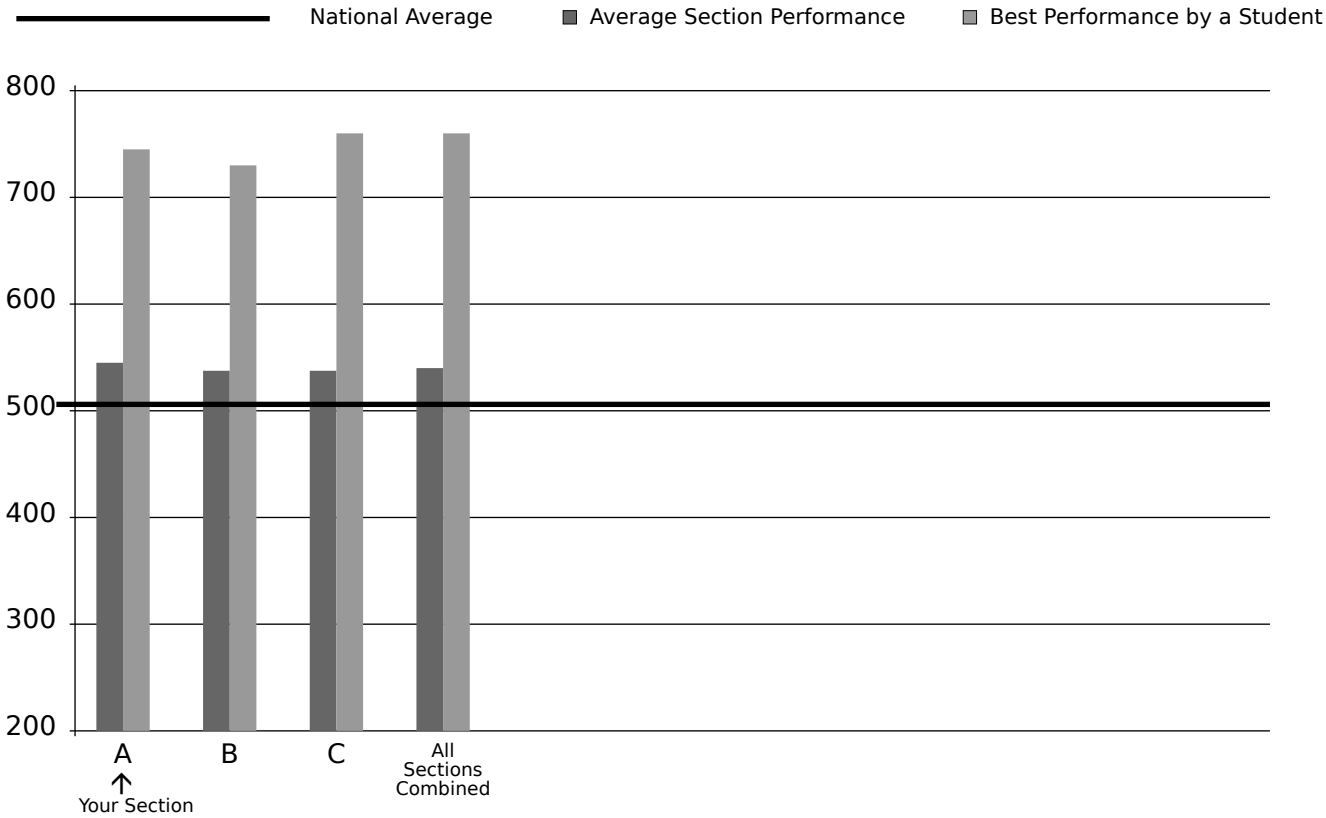
Performance as a percentage of National Average

Above Average(>105%) Below Average(<95%) Within Average(95% to 105%)

National average is taken as 100%. The value in the table indicate performance as compared to the national average.**E.g.** For a class average of 520 and overall average as 500, overall benchmarking %= (520/500)*100 = 104%. 95% to 105% is treated as similar to national average performance.Performance greater than 105% and below 95% is considered as performance above and below the national average range respectively.

Performance Data

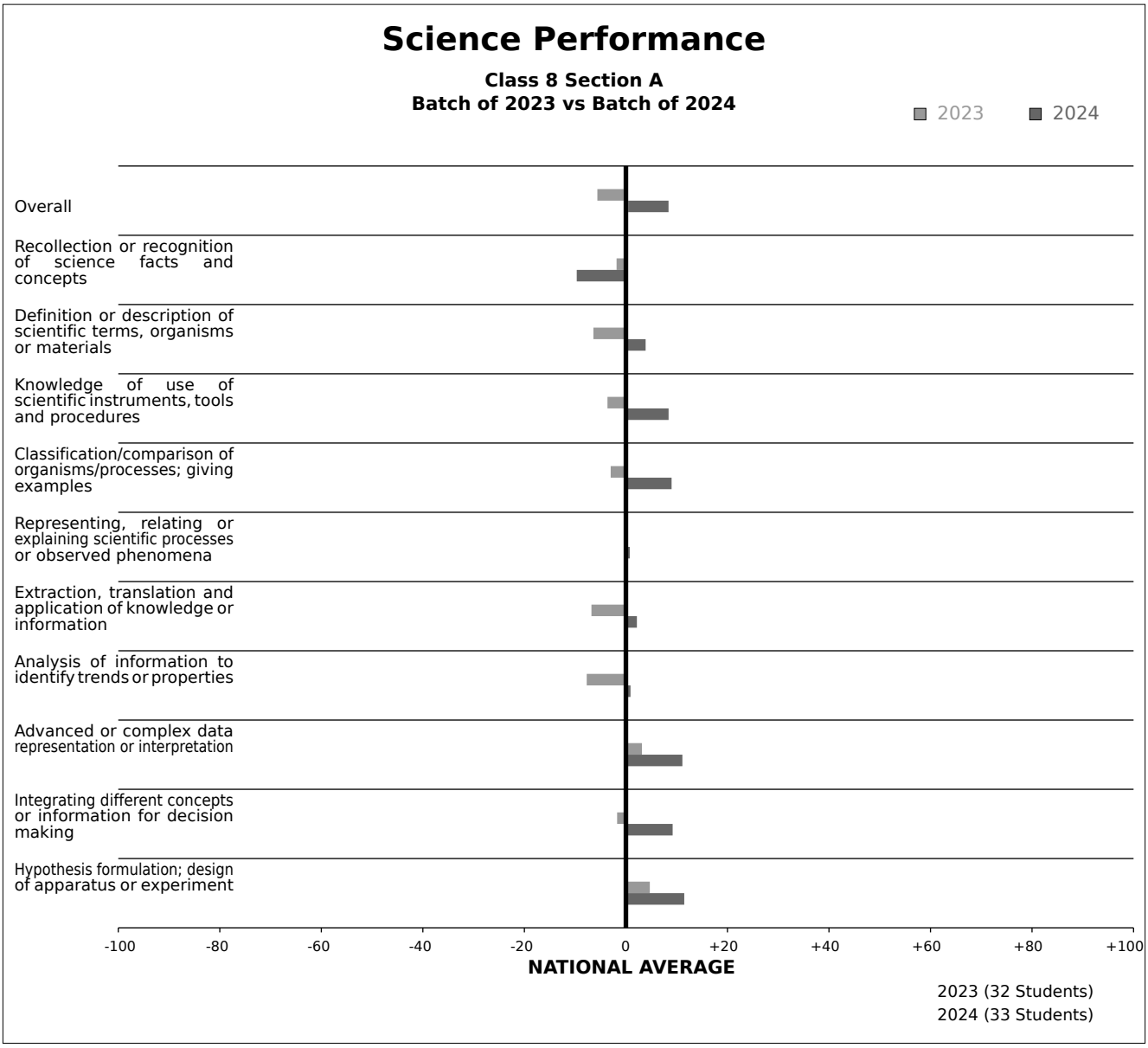
Comparison of section average and best performance with all schools in the country.



Comparison of performance of students of the current batch with the previous batch:

The following graph compares the performance of your current batch of students against the previous batch. The horizontal bars in the graph show how students of your class have fared against the national average in each skill across two years.

The thick black vertical line in the centre marks the national average.For example, a bar towards the right would mean that students performed higher than the national average that year in the particular skill.



This table shows how your students' performance compares to the overall school performance and the national average (all other schools in India). The skills with a '*' in the first column, if any, are those in which your students have performed better than the national average.

Class: 8 A Subject: Science Number of Students: 33

S.No.	Skill	Questions	Class 8 A Performance	Class 8 Performance (School)	Class 8 Performance (National)
			Average	Average	Average
1	Recollection or recognition of science facts and concepts	1,6,16,18,20	31.5%	35.7%	41.2%
2*	Definition or description of scientific terms, organisms or materials	3,17,31,38,44	47.9%	44.3%	44.0%
3*	Knowledge of use of scientific instruments, tools and procedures	7,13,22,43	47.7%	49.5%	39.3%
4*	Classification/comparison of organisms/ processes; giving examples	10,14,24,40	56.8%	55.1%	47.8%
5*	Representing, relating or explaining scientific processes or observed phenomena	2,8,34	36.4%	36.7%	35.6%
6*	Extraction, translation and application of knowledge or information	5,15,19,26,32,39	32.3%	31.3%	30.2%
7*	Analysis of information to identify trends or properties	4,12,25,33,36	43.6%	40.6%	42.7%
8*	Advanced or complex data representation or interpretation	29,30,35	61.6%	58.2%	50.5%
9*	Integrating different concepts or information for decision making	9,21,41,45	58.3%	57.4%	49.1%
10*	Hypothesis formulation; design of apparatus or experiment	11,23,27,28,37,42	56.6%	53.1%	45.1%

Note - The average performance is calculated from the number of correct responses given by students to the questions constituting specific skills.

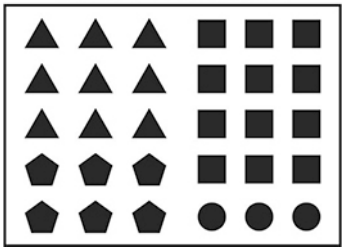
Strong skills		Weak skills	
1.	Advanced or complex data representation or interpretation	1.	Recollection or recognition of science facts and concepts
2.	Integrating different concepts or information for decision making	2.	Extraction, translation and application of knowledge or information

This part of the report shows questions that the children of your class found easy and difficult. This information can be used by you to focus your attention to specific questions and skills. You can also use the student performance table to focus on individual students. (Please refer to the ASSET paper for the complete text of these questions.)

Questions which students of your school found the MOST DIFFICULT compared to the rest of the country:

Q 4 Zubeda used four unique symbols to represent the students in her class as shown.

Students in class VIII



In Zubeda's class -

- 30% of the students come to the school on their cycles everyday.
- 50% of the students are girls.

Based on the given information, what does the

▲

symbol represent?

A boys who use a cycle to come to school
B girls who use a cycle to come to school
C boys who do not use a cycle to come to school
D (We cannot say for sure based on the given information.)

Skill: Analysis of information to identify trends or properties

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
4	38224	D	12%	44%	Your School	
					All Schools	

Q 6 What is the BEST time to watch vultures gliding in the sky?

A early morning **B** around midday **C** late evening **D** any time of the day

Skill: Recollection or recognition of science facts and concepts

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
6	38224	B	42%	58%	Your School	
					All Schools	

Q 17 The transport systems of both plants and animals involve water-based fluids.

Which of the following properties of water makes it suitable for this sort of a system?

A Water is an odourless substance.
B Many substances dissolve in water.
C Water exists in three physical states.
D Oxygen is a part of the water molecule.

Skill: Definition or description of scientific terms, organisms or materials

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph
17	38224	B	30%	45%	Your School / All Schools

Questions which students of your school found the EASIEST compared to the rest of the country:

Q 13 One evening, four friends were chatting in a park.

Anil: There are so many Mango trees here.
Bhavya: I bet there are many birds in the park.
Charita: There are probably many mosquitoes here.
Danny: My doctor told me that mosquito bites can cause Malaria.

Who has made a statement that a scientist would call an OBSERVATION?

A Anil
B Bhavya
C Charita
D Danny

Skill: Knowledge of use of scientific instruments, tools and procedures

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph
13	38224	A	64%	27%	Your School / All Schools

Q 35 The graphs show the number of people entering three shopping malls P, Q and R.

The number of people entering which of the malls is likely to increase the MOST if the numbers are recorded for some more time?

A shopping mall P
B shopping mall Q
C shopping mall R
D (All the three malls will show an equal increase.)

Skill: Advanced or complex data representation or interpretation

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
35	38224	B	67%	39%	Your School	
					All Schools	

Q 33 The cups and boxes shown here are made of a plastic called styrofoam.

When used and thrown into waste they cause problems to the environment because they do not get decomposed.

A company collects used and discarded styrofoam and makes it into a material called "styrowood". It is waterproof and ideal for making fences and outdoor furniture which are usually made out of wood.

The making and use of "styrowood" will DIRECTLY result in _____.

A conservation of resources
B discovery of resources
C depletion of resources
D pollution of resources

Skill: Analysis of information to identify trends or properties

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
33	38224	A	85%	62%	Your School	
					All Schools	

From the questions that students across the country have found the most difficult, we have chosen a few to highlight potential misconceptions that students might have. For each question, we have given a break-up (in percentage form) of the number of students choosing a particular option. This has been done both for the section of your school and all the students nationally. Compare these to find out how your students have performed.

We have highlighted in light grey the correct option and in dark grey the most common wrong answer, and provided an explanation of why students are choosing the most common wrong option. It is recommended that you discuss these questions with the students to uncover students' rationale in choosing the most common wrong option. (Please refer to the ASSET paper for the complete text of these questions.)

Q 15 Pollination is a step in the process of sexual reproduction in plants. It can take place in three ways.

P) A male gamete from a flower fuses with the female gamete of the same flower to form a seed.
Q) A male gamete from a flower fuses with the female gamete of another flower on the same plant to form a seed.
R) A male gamete from a flower fuses with the female gamete of a flower on another plant (of the same species) to form a seed.

Planting a seed produced by which of the methods of pollination is likely to produce a plant that has characteristics IDENTICAL to the parent plant?

A only P **B** only P and Q **C** all, P, Q and R **D** none of P, Q and R

The question tests students' understanding that ALL the three methods of pollination described involve sexual reproduction. Students choosing the correct OPTION D understand that sexual reproduction leads TO the mixing of gametes LEADING TO the formation of characteristics IN the offspring that are NOT exactly the same AS those IN the parent plant. Students choosing OPTION A might be thinking that since gametes FROM the same flower are involved, the characteristics transmitted will be the same AS the parent plant. Students choosing OPTION B might be thinking that since the flowers ON the same plant are involved, the characteristics transmitted will be the same AS the parent plant. Students choosing OPTION C might be thinking that since the flowers ON the same species are involved, the characteristics transmitted will be the same AS the parent plant.

Papercode: 38224, Question no: 15					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	27%	52%	21%	0%	33
National	28%	32%	31%	8%	11685

Q 1 The liver secretes bile, needed to digest fat in our food. The pancreas secretes several enzymes and hormones needed to break down food.

Which of the following is true of the food that we eat?

A It passes only through our liver.
B It passes only though our pancreas.
C It passes through both our liver and pancreas.
D It passes neither through our liver nor pancreas.

This question was designed to test whether students know that food does not pass through our liver and pancreas. The most common wrong answer indicates that students feel that all the food passes through both the liver and the pancreas! The process of digestion and the path of ingested food should be discussed among students to see what other notions they may have.

Papercode: 38224, Question no: 1					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	9%	18%	42%	27%	33
National	9%	11%	38%	41%	11685

Q 16 Plants produce food in the form of carbohydrates in the leaves during the process of photosynthesis.

Which of the following normally happens to the food produced?

i	It is broken down into components that are used in the synthesis of proteins and fats.
ii	Extra food produced is stored as starch in different parts of the plant.
iii	It is utilised by the cells of the plant during respiration to produce energy.

A only i and ii **B** only i and iii **C** only ii and iii **D** all - i, ii and iii

This question was designed TO test IF students understand what happens TO the food produced BY a plant. Students are familiar WITH the concept that the glucose (carbohydrate) produced BY a plant during photosynthesis IS utilized BY the cells TO produce energy BY respiration, AND that excess glucose IS converted TO starch which IS stored IN different plant parts, therefore they may choose OPTION C. Students should know that besides carbohydrates, a plant also requires proteins AND fats. They should understand that plants also synthesize other substances containing carbon, hydrogen AND oxygen LIKE proteins AND fats. The carbon, hydrogen AND oxygen used TO make proteins AND fats comes FROM the breakdown of the carbohydrate, whereas the nitrogen IN proteins comes FROM soil nutrients absorbed BY the roots.

Papercode: 38224, Question no: 16					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	27%	9%	36%	24%	33
National	17%	17%	30%	35%	11685

This section is crafted to aid students in practicing weaker skills. Use these in class discussions, gauge common understanding gaps, and tailor lessons. Answers and explanations are included. Employ a mix of question types to stimulate students' thinking. Here's how we've tailored the practice questions for each subject: For English, if reading comprehension is a challenge, we've provided full passages to work on. In Math, Science, and Social Studies, we've chosen questions based on areas where you've shown room for growth. Hindi practice focuses on non-reading comprehension skills. Computational Thinking covers a variety of skills across the class.

Skill: Recollection or recognition of science facts and concepts (32%)

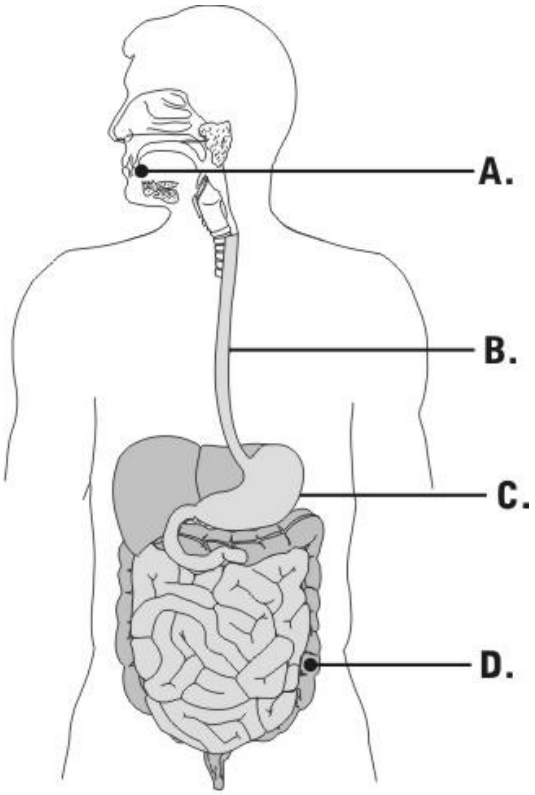
Q 1 Which of these molecules is composed of EXACTLY 5 atoms?

A $NaHCO_3$
B $Ba(OH)_2$
C K_2SO_4
D N_2O_5

Explanation:

B: Symbols of all elements are written with capital letters. To distinguish some elements from others, 2 letters might be used, but even for them, the first letter is in capital. Na, H, C, O, etc. are the symbols of different elements. Na represents a single element, sodium, and it does not mean 2 elements, N and a. $Ba(OH)_2$ is the molecule that has 5 atoms - 1 Ba atom, 2 H atoms and 2 O atoms.

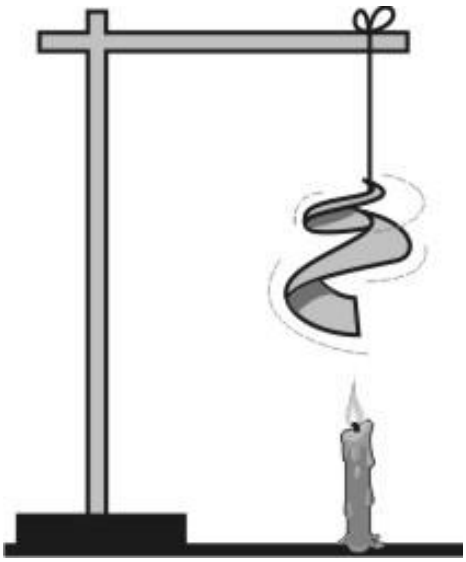
Q 2 In which of these organs of the digestive system does food get digested due to the presence of acids?



Explanation:

C: Food is mixed with acids in the stomach, which is marked by part C. Part D is the large intestine, in which most of the water gets reabsorbed from the undigested food.

Q 3 A paper strip spiral was hung from a stand in a closed room as shown below. A candle placed below it was lighted. As the candle started burning, the paper moved because of the movement of air. This movement occurs due to _____.



A radiation **B** convection **C** conduction **D** evaporation

Explanation:

B: In the setup shown, the candle warms the air above it, which rises up and then moves the paper strip. It is the movement of the warmed matter, ie.air, that causes the movement. This mode of heat transfer is called convection. Radiation is a mode of heat transfer which does not require the movement of matter. Electromagnetic waves are transmitted through space, by radiation.

Q 4 Four friends were discussing whether the rusting of an iron nail is a physical or a chemical change. Here is what each one said.

Sarla: 'It is a chemical change because a new substance 'rust' is formed in the process.'
Ravi: 'It is a physical change because I can scrape off the rust and the original nail is got.'
Shyam: 'It is a physical change because it happens without the application of heat.'
Mariam: 'It is a chemical change because the rust that is formed looks different.'

Whose statement is correct?

A Sarla **B** Ravi **C** Shyam **D** Mariam

Explanation:

A: A chemical change is a process in which a new substance is formed, whose properties are not the same as the original substance. When iron rusts, it forms a new substance, and its properties are not the same as iron. Hence, rusting is a chemical change. Even if we scrape off the rust, its properties do not change, and they will be different from iron.

Q 5 "When light is reflected from a smooth surface, the angle of incidence must be equal to the angle of reflection."

Under which of the following situations is the above statement true?

A It is always true.
B It is true only if the surface is CONCAVE or PLANE.
C It is true only if the surface is made of GLASS.
D It is true only if the surface is PLANE.

Explanation:

A: Light reflects from all surfaces in a similar manner, following the law of reflection. The surface may be plane or curved, but the angle that the incident light ray makes with the normal at the point of incidence is the same as the angle that the reflected ray makes with it.

Q 6 A hydro carbon C_4H_{10} has a boiling point of 0^0C. Assuming it is stable at 25^oC, what would be its state at that temperature(all other conditions like pressure remaining the same)?

A Solid
B Liquid
C Gas
D (Cannot predict without knowing the melting point)

Explanation:

C: Boiling means change of liquid state to gaseous state. The hydrocarbon has boiling point of 0° C. It means that at 0^oC it gets converted into gaseous state and will be found in that state at all temperatures higher than 0^oC. So at 25^oC it will be found in its gaseous state.

Q 7 Three gases found in Earth's atmosphere are carbon dioxide, nitrogen and oxygen. What is their order of abundance from greatest to least?

A nitrogen, oxygen, carbon dioxide
B nitrogen, carbon dioxide, oxygen
C oxygen, nitrogen, carbon dioxide
D carbon dioxide, oxygen, nitrogen

Explanation:

A: Nitrogen forms 78% of the air, whereas, oxygen and carbon dioxide form 21% and 0.03% of the air, respectively. Hence, their order of abundance from the greatest to the least is nitrogen, oxygen and carbon dioxide.

Q 8 Which of the following body parts is most likely to help a mammal in camouflaging itself from its enemies?

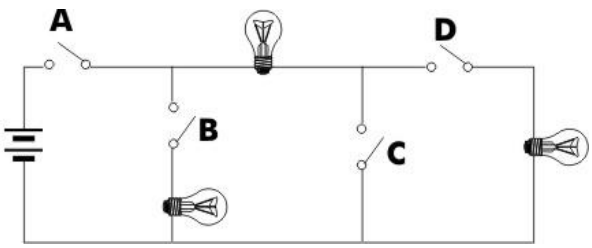
A Eyes
B Nails
C Heart
D Fur

Explanation:

D: Camouflage is a method by which an otherwise visible animal becomes invisible because its appearance matches with the surrounding environment. It helps to protect the animals from other animals who might eat them, and even helps the animals hide so that they cannot be recognised by their prey. Eyes and nails are too small and so, even if they match the surroundings, the organism will be easily visible. Skin and fur cover most of the animal's body and so, if that matches the surroundings, the animal will be almost invisible.

Skill: Extraction, translation and application of knowledge or information (32%)

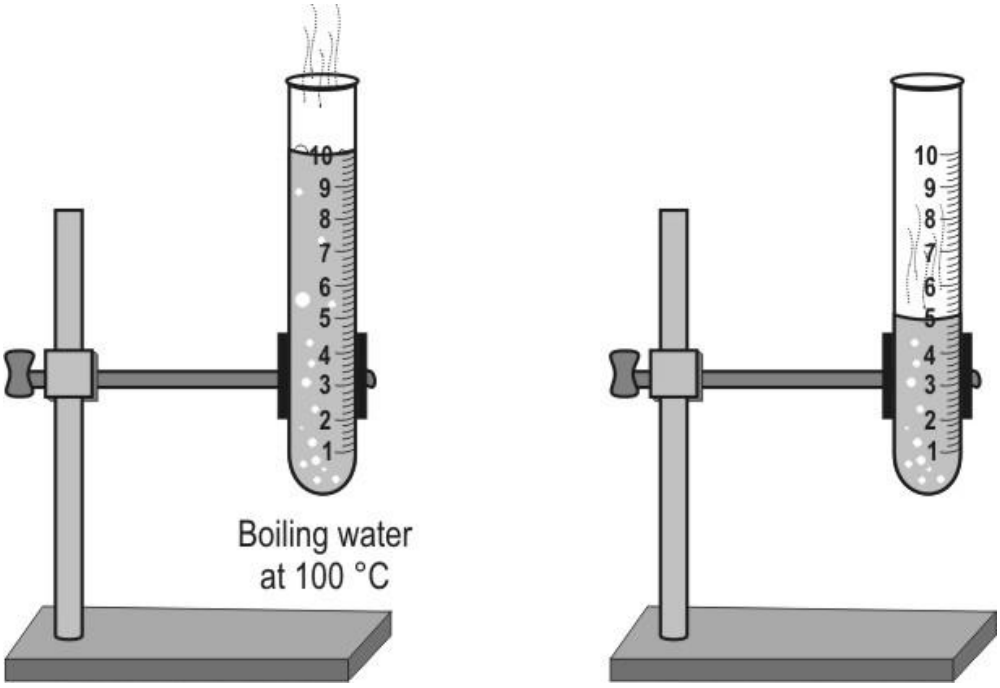
Q 1 Which switch A, B, C or D MUST be open to ensure that no bulb will glow?



Explanation:

A: Look carefully at the given circuit. Let us imagine each switch being open at a time. If switch B is open, 2 bulbs will still glow. If switch C is open, all 3 bulbs will glow, because C is not needed to complete the circuit between the battery and the bulbs. If switch D is open, 2 bulbs will glow - the bulb next to B and the one before D. However, if switch A is open, the wire leading to the end of the battery will be disconnected, and hence the circuit will not receive any power. Thus, option A is the only correct answer.

Q 2



Shakira heated 10 ml of water in a test-tube till it started boiling. She noted that the temperature of the water was 100°C. If she pours out half the water into the sink, what will be the approximate temperature of the water left in the test-tube immediately after this?

A 30°C
B 50°C
C 100°C
D 200°C

Explanation:

C: Water boils at 100°C. This is true for any amount of water. If you take boiling wate and quickly pour out some of it, and then immediately measure its temperature, it will still be 100°C. This is because heat takes some time to escape from the water. If water is boiling at 100°C, any part of the water will be 100°C, even if some water is taken out of it. Thus, only option C is correct.

Q 3 The human body (especially when dry) is a poor conductor of electricity, yet we get an electric shock because it does conduct some electricity. Which of these is a similar example when a poor conductor allows electricity to pass through it?

A During a thunderstorm, lightning strikes when electric charge passes through the air, which is normally a poor conductor.
B Electricity passes through the copper wires inside the motor of a ceiling fan as copper is a good conductor of electricity.
C Electricity passes through the metallic filament in a bulb when it glows, though the filament is normally a poor conductor.
D Electricity passes through the walls of a building when a light is turned on, though bricks are normally poor conductors.

Explanation:

A: Lightning is the result of a very heavy build up of opposite charges between the clouds and the earth's surface or between clouds. This build-up of charges results in a sudden discharge of electricity, even through a poor conductor like air. Copper wires and metallic filaments are not poor conductors of electricity. The electricity for lights comes through metal wires, and not directly through the bricks, hence, option D is incorrect.

Q 4 You know that unlike magnetic poles attract - the south pole of one magnet is attracted to the north pole of another magnet within its magnetic field.

A magnetic compass responds to the earth's magnetic field and comes to rest in the North-South orientation with the North pole of the compass pointing to the Earth's geographic North.

What does this imply?

A The earth's magnetic South pole is near its geographic North pole.
B The earth's magnetic North pole is near its geographic North pole.
C The earth's magnetic poles are perpendicular to its geographic poles.
D The earth's magnetic poles lie at the core of the earth, which is a mass of molten iron.

Explanation:

A: The magnetic poles of a magnet are named according to the direction they point in. The north poles of all magnets point to the North direction. We know that unlike magnetic poles attract. The reason magnets align in a certain direction on earth, is because there is a giant magnet within the earth. The South pole of the earth's magnet is towards the geographic North pole. This is the reason it can attract the North pole of other magnets. The earth's magnetic poles cannot be perpendicular to the geographic poles because the other magnets would, then, also be aligned perpendicular to the geographic poles, which does not happen. If the earth's magnetic North pole was near its geographic North pole, then the South poles of all other magnets would point to the earth's geographic North pole!

Q 5 Sap is the name given to the liquids transported by xylem and phloem. What does it contain?
P. products of photosynthesis
Q. chlorophyll from the leaves

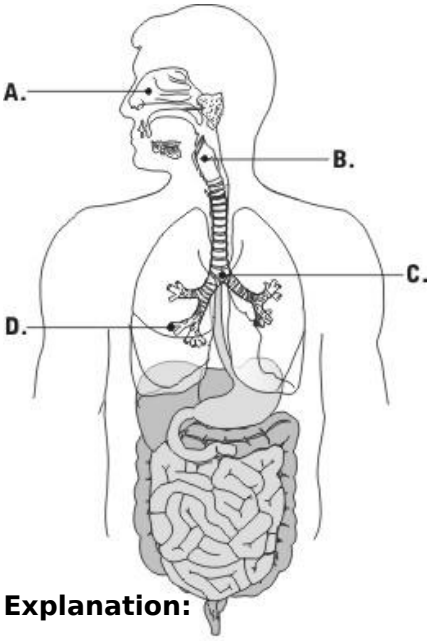
R. water and minerals from the soil
S. waste fluids excreted by the plants

A P and R **B** Q and R **C** Q and S **D** P and S

Explanation:

A: The xylem mainly transports water and minerals, and the phloem mainly transports the food produced by photosynthesis. Chlorophyll is never transported from the leaves, and the waste fluids are usually given out by the plant organs themselves. Thus, it is mainly P and R that are found in the xylem and phloem.

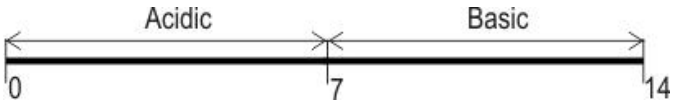
Q 6 A valve is a structure in a hollow organ that controls the direction of the flow of matter through it. Shown here is a schematic diagram of the respiratory and the digestive system. At which of the positions are you likely to find a valve?



Explanation:

B: Look at the labelled parts in the diagram. Point A is the nose, which does not require a valve. Point C is the division between the wind-pipe and the lungs, which does not require a valve either. Point D is the part where the branches of the bronchi divide into the lungs - they also, do not require a valve. Only point B, which has the openings of the respiratory and the digestive system, need a valve so that food does not enter the wind-pipe and choke us.

Q 7 Shown below is the range of pH values and its significance.



Chemical	pH
Hydrochloric acid	2
Ammonium hydroxide	10
Sulphuric acid	1
Acetic acid	5

The strongest acid among the substances listed above is

A Acetic acid
B sulphuric acid
C ammonium hydroxide
D hydrochloric acid

Explanation:

B: The pH scale in the first figure shows that acidic substances lie within the pH range between 0 to 7. Basic substances lie between 7 and 14. This means that as the pH range goes from 0 to 14, the acidity decreases and the basicity increases. The opposite is true as the pH decreases from 14 to 0. Thus, the substance with the lowest pH will be the strongest acid. In this case, it is sulphuric acid, option B.

Q 8 Farmer John grew a variety of plants on his farm. For some of his plants he always preferred to use the asexual mode of reproduction rather than the sexual mode of reproduction. Which of these reasons would explain why he did this?
The resulting plants always _____.

A produced bigger and better fruits than the parent plant
B had characteristics identical to the parent plant
C produced more seeds than the parent plants
D had only the best characteristics of the parent plants

Explanation:

B: Plants that grow as a result of asexual reproduction will have characteristics identical to their parents because they grow from an older part of the parent plant, and not a new seed. They will not have bigger or better fruits, since the plant will have characteristics that are identical to the parent plant. A plant that grows from a new seed, formed due to sexual reproduction, will not be identical to the parent plants. Hence, only option B is correct.

Notes